INDIAN STATISTICAL INSTITUTE, KOLKATA

ECONOMIC STATISTICS PROJECT

Demographic Analysis and Comparative Study of Maharashtra and Uttar Pradesh

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Project Report

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1 Introduction:

1.1 What is already known?

India shows large variations in socio-economic status and health indicators across states. NFHS reports consistently highlight differences in education, fertility patterns, maternal and child health, and use of healthcare services. States with stronger socio-economic development generally show better health outcomes and lower fertility. These differences make interstate comparison important for understanding demographic and health inequalities.

1.2 Research gap:

Although several NFHS-based reports describe state-level health indicators, comparative analyses linking socio-economic characteristics with fertility and maternal–child health outcomes between high-performing and low-performing states are limited, especially using micro-data from NFHS-5. Few studies simultaneously examine multiple dimensions (fertility, early childbirth, contraception, and child health) within the same analytical framework.

1.3 Aim of the present study:

To compare socio-economic and demographic determinants of fertility and maternal–child health outcomes between Uttar Pradesh and Maharashtra using NFHS-5 data, and to examine how differences in education, wealth, contraception, and vaccination may contribute to variations in fertility and early childbirth between the two states.

2 Research Questions:

- Do education, wealth, and contraception explain observed differences in fertility between Uttar Pradesh and Maharashtra?"
- How does the prevalence of early childbirth (first birth before age 20) differ between Uttar Pradesh and Maharashtra?
- After birth within 3 days if food other than breast milk is given like (honey, glucose water,fruit juice,JANAM GHUTTI) then it can increase infant mortality rate.(Ref.Col 1269-1376)
- How does fertility vary by place of residence (urban vs rural)?Can compare between two states.
- While both Maharashtra and UP have an income gradient from west to east, signifying high spatial variance of income, we want to know that for which state is this variance higher?

3 Data sources:

For Research Questions 1, 2, 3 and 4 we have used **NFHS-5 Dataset** and for Research Question 5 we have referrred from **Economic Survey of Maharashtra 2021-22** and **Statistical Diary, U.P. 2021**.

4 Research Question 1:

4.1 Research Objectives:

The study aims to examine whether socio-economic factors explain the variation in fertility levels. Specifically, we test whether a woman's education level, household wealth, and contraception use have a statistically significant effect on fertility.

4.2 Research Hypothesis:

Let the regression model be:

$$Y = exp(\beta_0 + \beta_{\text{educ}}(\text{Education}) + \beta_{\text{wealth}}(\text{Wealth}) + \beta_{\text{contraception}}(\text{Contraception})) + \varepsilon$$

where Y denotes the number of children ever born.

Null Hypothesis (H_0):

$$H_0 : \beta_{\text{educ}} = \beta_{\text{wealth}} = \beta_{\text{contraception}} = 0$$

There is no significant association between education level, wealth index, contraception use and fertility.

Alternative Hypothesis (H_1):

$$H_1 : \beta_{\text{educ}} \neq 0 \quad \text{or} \quad \beta_{\text{wealth}} \neq 0 \quad \text{or} \quad \beta_{\text{contraception}} \neq 0$$

At least one of the factors (education, wealth, or contraception use) has a significant association with fertility.

4.3 Methodology

Survey Design

The analysis uses data from the National Family Health Survey (NFHS–5), which follows a complex survey design. To ensure unbiased and representative estimates, the survey design information was incorporated using the `survey` package in R. Three elements of the design were applied:

- **Sampling weights** ($V005$) were used to adjust for unequal probability of selection.
- **Stratification** ($V022$) was applied to account for the stratified sampling method used across districts and urban/rural areas.
- **Cluster sampling** ($V021$) was used to adjust for respondent clustering within primary sampling units (PSUs).

The survey design was specified in R as:

```
svydesign(ids = 021, strata = 022, weights = Ṽ005, data = dataset)
```

Poisson Survey Regression Model

Fertility (measured as number of children ever born, variable $V201$) is a count variable. Hence, a **Poisson regression model** was used. Since NFHS uses a complex survey design, ordinary Poisson regression would underestimate standard errors. Therefore, a **survey–adjusted Poisson regression** (`svyglm`) was applied.

The model estimated is:

$$\log(E[Y_i]) = \beta_0 + \beta_1 \text{Education}_i + \beta_2 \text{Wealth}_i + \beta_3 \text{Contraception}_i$$

Where:

- Y_i = fertility (children ever born) for woman i
- Education ($V106$) = categorical (no education, primary, secondary, higher) Wealth ($V190$) - wealth index
- Contraception ($V313$) = categorical (no method, traditional, modern)

After fitting the model separately for Maharashtra and Uttar Pradesh, the estimated coefficients were compared. Exponentiating the coefficients gives **Incidence Rate Ratios (IRRs)**:

$$IRR = e^{\beta}$$

Values of $IRR < 1$ imply a reduction in fertility relative to the reference category.

Variable Description

Dependent Variables

- Number of children ever born (`v201`)

Independent Variables

Three socio-economic and behavioural predictors were included in the model:

1. **Education Level of the Woman** ($V106$)

Value	Label
0	No education
1	Primary
2	Secondary
3	Higher

2. **Wealth Index** ($V190$) (constructed by NFHS using PCA of household assets)

Value	Label
1	Poorest
2	Poorer
3	Middle
4	Richer
5	Richest

3. **Current Contraceptive Use** ($V313$)

Value	Label
0	No method
1	Folkloric method
2	Traditional method
3	Modern method

Each variable was treated as categorical in the Poisson survey regression model, and dummy coding was used, taking the lowest category as reference.

Assumption:

Poisson regression relies on the following assumptions:

1. The dependent variable is a count variable and takes non-negative integer values.
2. The response variable follows a Poisson distribution:

$$Y_i \sim \text{Poisson}(\lambda_i)$$

3. The mean and variance of the response are equal (equidispersion):

$$E(Y_i) = Var(Y_i) = \lambda_i$$

4. The logarithm of the expected count is a linear combination of the predictors:

$$\log(\lambda_i) = X_i\beta$$

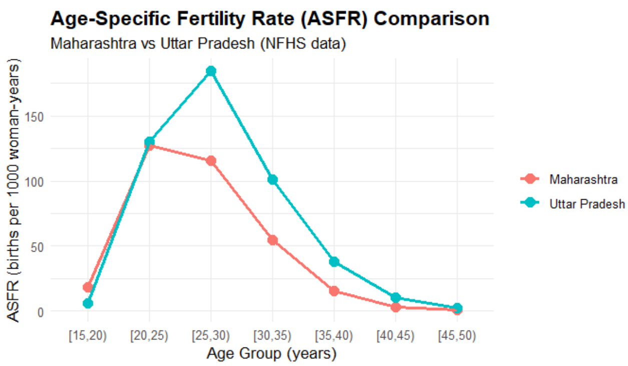
5. Observations are independent. In our case, NFHS uses a complex survey design, so clustering, stratification, and sampling weights were corrected using survey-weighted Poisson regression.
6. There is no perfect multicollinearity among predictors.

4.4 Salient Findings:

Descriptive Statistics

We first compare the fertility patterns of Uttar Pradesh and Maharashtra using Total Fertility Rate (TFR) and Age-Specific Fertility Rates (ASFR). We can observe Uttar Pradesh has higher total fertility rate than Maharashtra.

The ASFR pattern shows that both states have almost identical fertility levels in the 15–19 age group. Maharashtra reaches its peak fertility at age 20, whereas Uttar Pradesh reaches its peak at age 25. After age 20, Uttar Pradesh consistently shows higher ASFR values than Maharashtra, indicating that fertility remains higher for longer ages in Uttar Pradesh.



Poisson Survey Regression Results

The mean of Y variable is 1.69 and the variance is 1.90. Hence poisson regression model is applicable.

Table 1 presents the results of the Poisson survey-weighted regression models estimating the expected number of children ever born (`V201`) as a function of mother's education (`V106`), household wealth index (`V190`), and current use of contraception (`V313`). Categorical variables are treated as factors, with the lowest category taken as reference (No education for schooling, Poorest for wealth, and No method for contraception).

Table 1: Survey-weighted Poisson regression results for Uttar Pradesh and Maharashtra

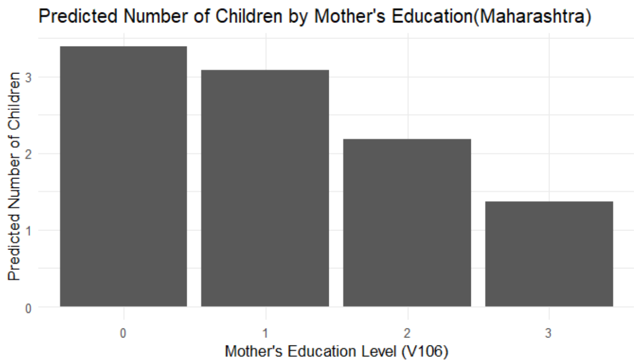
Variable	**UP: Estimate (SE)**	**MH: Estimate (SE)**
Intercept	0.7403 (0.0100)*	0.3303 (0.0250)*
Education (ref = No education)		
Primary	−0.3152 (0.0098)*	−0.0976 (0.0182)*
Secondary	−0.8640 (0.0091)*	−0.4389 (0.0172)*
Higher	−1.1854 (0.0129)*	−0.9059 (0.0343)*
Wealth (ref = Poorest)		
Poorer	−0.0493 (0.0076)*	0.0063 (0.0211)
Middle	−0.0367 (0.0087)*	−0.0107 (0.0205)
Richer	−0.0200 (0.0098)*	−0.0226 (0.0198)
Richest	0.0179 (0.0115)	−0.0512 (0.0228)*
Contraception (ref = No method)		
Folkloric	0.3950 (0.0112)*	—
Traditional	0.7755 (0.0100)*	0.7721 (0.0381)*
Modern	0.7686 (0.0088)*	0.9031 (0.0195)*

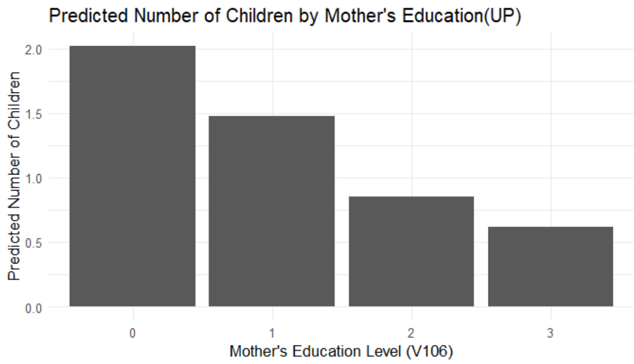
Note: * $p < 0.001$, ** $p < 0.01$, * $p < 0.05$. Contraception category "folkloric" does not appear in Maharashtra due to negligible sample.

Interpretation. For both states, the coefficients for increasing education levels are negative and statistically significant, indicating that fertility decreases monotonically as education increases. Wealth shows a weak association in Maharashtra but a stronger, negative association in Uttar Pradesh. Surprisingly, contraception use is associated with *higher* fertility in both states, which may indicate reverse causality: women tend to adopt contraception *after* achieving desired parity. Furthermore,

- **Fertility Pattern (ASFR & TFR)**
  - Uttar Pradesh consistently exhibits higher fertility compared to Maharashtra across all age groups.
  - The Age-Specific Fertility Rate (ASFR) curve shows the typical rise–peak–decline pattern in both states.
  - Maharashtra reaches its fertility peak in the **20–24** age group, indicating early and concentrated childbearing.
  - Uttar Pradesh peaks later in the **25–29** age group, and ASFR remains high even beyond 30 years.
  - After age 20, **Uttar Pradesh dominates the ASFR curve** across all subsequent age groups.
  - Total Fertility Rate (TFR) is higher in Uttar Pradesh than in Maharashtra, reflecting the broader reproductive span in UP.
  - Maharashtra shows a steeper decline in ASFR after age 30, suggesting better contraception use and spacing.
- **Regression Model Findings (Poisson Survey Regression)**
  - Education has a strong negative effect on fertility in both states: as education moves from "No schooling" to "Higher education", expected number of children decreases significantly.

- The fertility reduction due to education is **more pronounced in Uttar Pradesh** than in Maharashtra.
- Wealth index:
  * In Uttar Pradesh, higher wealth levels are significantly associated with lower fertility.
  * In Maharashtra, the effect of wealth is weak and mostly statistically insignificant.
- Contraception use shows a positive coefficient in both states, indicating adoption of contraception **after reaching the desired family size** (reverse causality effect).
- The magnitude of the effect for modern methods is stronger in Maharashtra than in Uttar Pradesh.

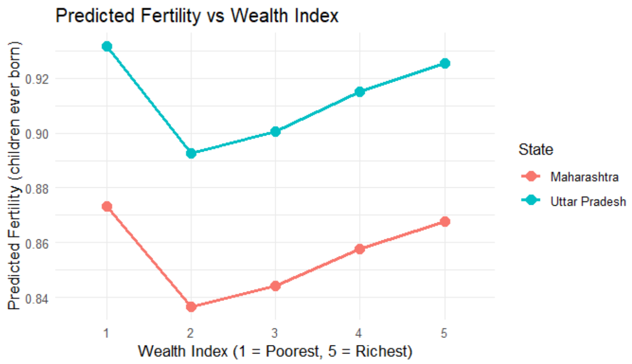


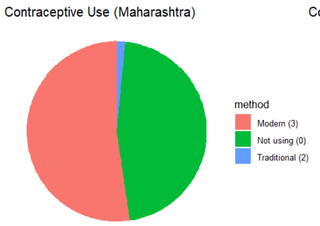


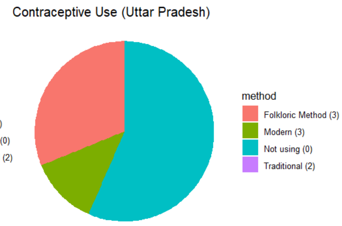
- **Demographic and Behavioural Insights**
  - Fertility is more concentrated in the 20–29 age bracket in Maharashtra, implying planned family formation.
  - Uttar Pradesh shows fertility spread over a wider reproductive age range, indicating delayed spacing or low family planning uptake.
  - Education and wealth reduce fertility more effectively in Uttar Pradesh, suggesting state-level differences in socioeconomic influence.
  - Modern contraception use is higher in Maharashtra, while Uttar Pradesh shows greater use of traditional or temporary spacing methods.

4.5 Conclusion:

Even after controlling for education, wealth, and contraception, fertility remains significantly higher in Uttar Pradesh than in Maharashtra. Differences stem from peak fertility age, socioeconomic factors, and contraception behavior.







5 Research Question 2:

5.1 Research Objectives:

Broad Objective:

To compare the prevalence of early childbirth (first birth before age 20) among women in Uttar Pradesh and Maharashtra.

Specific Objectives:

1. To estimate the proportion of women who had their first birth before age 20 in each state.
2. To examine the socio-demographic factors (such as education and residence) associated with early childbirth in Uttar Pradesh and Maharashtra.

5.2 Research hypothesis

Null Hypothesis $(H_0): \ p_{\text{Maharashtra}} = p_{\text{Uttar Pradesh}}$

Alternative Hypothesis $(H_1): \ p_{\text{Maharashtra}}$ and $p_{\text{Uttar Pradesh}}$ are significantly different.

Here, $p_{\text{Maharashtra}}$ and $p_{\text{Uttar Pradesh}}$ denote the weighted prevalence of early childbirth (first birth before age 20) among women in Maharashtra and Uttar Pradesh, respectively, calculated using the `svyby()` function in R.

5.3 Methodology

Sampling weights provided in the dataset (`V005`) were applied to ensure nationally representative and unbiased estimates.

t-test

A weighted two-sample t-test was used to compare the prevalence of early childbirth between the two states. The test was based on the null hypothesis that the prevalence of early childbirth in Maharashtra and Uttar

Pradesh is equal, against the alternative that they are significantly different. The decision criterion was based on the p-value: if $p < 0.05$, the null hypothesis was rejected, indicating a statistically significant difference in prevalence between the two states.

Logistic Regression Model

Since the dependent variable `early_birth` is binary (1 = first birth before age 20, 0 = otherwise), a logistic regression model was used to identify socio-demographic factors associated with early childbirth. Logistic regression is appropriate for binary outcomes as it models the log-odds of the probability of the event. Moreover, because NFHS follows a complex sampling design, the model was estimated using a survey-adjusted logistic regression (`svyglm`) to obtain correct standard errors.

The model estimated is:

$$\log\left(\frac{P(Y_i = 1)}{1 - P(Y_i = 1)}\right) = \beta_0 + \beta_1 \text{State}_i + \beta_2 \text{Residence}_i + \beta_3 \text{Education}_i + \beta_4 \text{Wealth}_i + \beta_5 \text{Religion}_i + \beta_6 \text{Caste}_i + \varepsilon_i$$

Where:

- Dependent Variable: Y_i = early childbirth status for woman i (1 = first birth before age 20, 0 = otherwise)
- Independent Variables: **State (V024)**: 9 = Uttar Pradesh, 27 = Maharashtra, **Education level (V106)**, **Wealth Index (V190)**, **Residence (V025)**: Urban/Rural, **Religion (V130)**, **Caste (V131)**

After estimating the model, the coefficients (β) were exponentiated to obtain Odds Ratios (OR = e^{β}). Values of OR greater than 1 indicate higher odds of early childbirth relative to the reference category, while values less than 1 indicate lower odds.

5.4 Salient Findings

Graphs and Their Interpretation

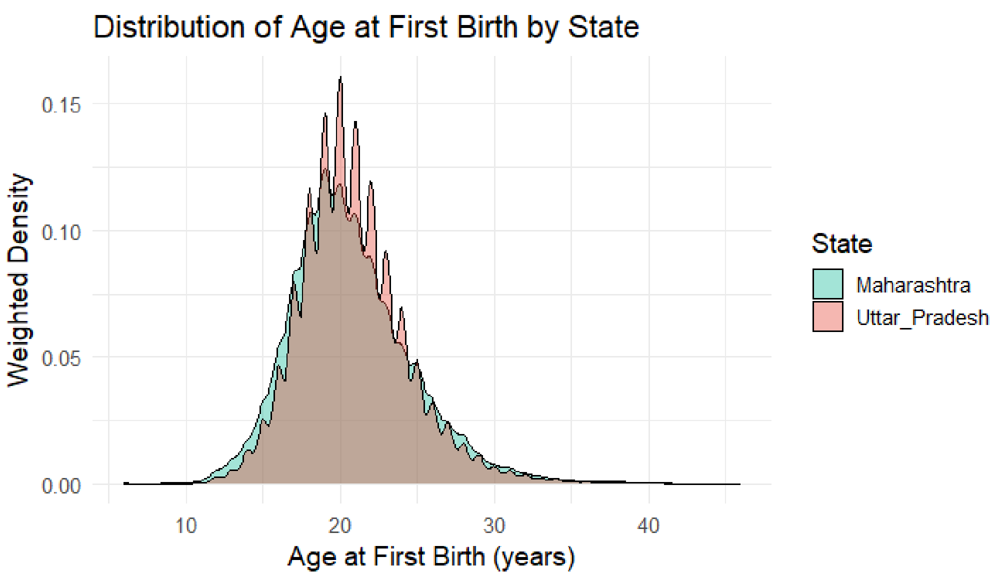


Figure 1

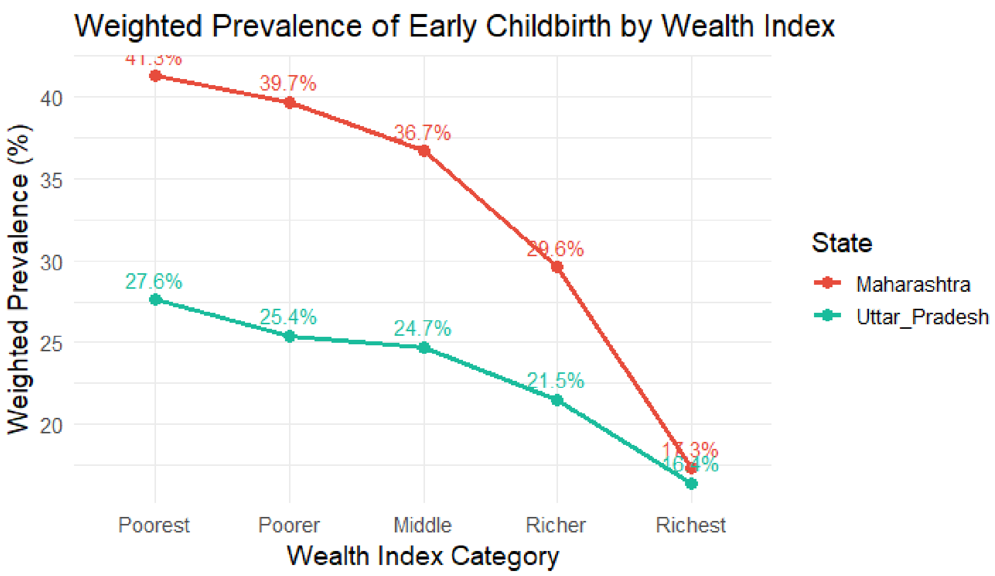


Figure 2

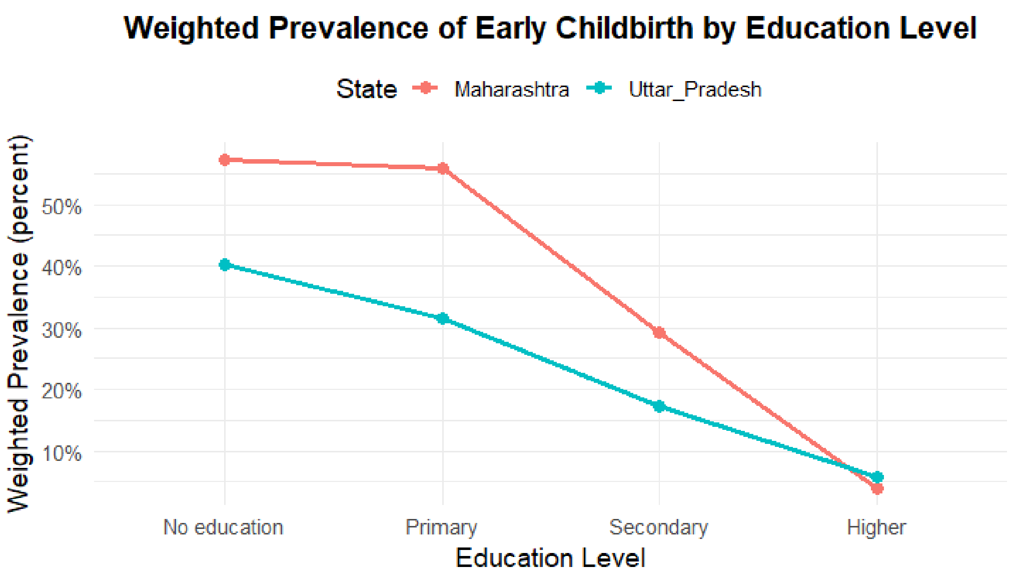


Figure 3

From Figures 2 and 3, it can be observed that the prevalence of early childbirth declines sharply with increasing levels of education and wealth index in both Maharashtra and Uttar Pradesh. Maharashtra consistently shows a higher prevalence all groups, though it exhibits a steeper rate of decline in both cases. Moreover, for women with higher education, Maharashtra's prevalence falls slightly below that of Uttar Pradesh, indicating that as education increases, Maharashtra's relative advantage becomes more evident.

A similar trend is observed when the data are grouped by residence. In Maharashtra, the weighted prevalence of early childbirth is **23.4% in urban areas** and **36.0% in rural areas**, whereas in Uttar Pradesh, it is **20.2% in urban areas** and **24.6% in rural areas**.

t-test Findings

The weighted prevalence of early childbirth in **Maharashtra** was **0.4190**, while in **Uttar Pradesh** it was **0.3703**.

The estimated difference in means was **-0.0487**, indicating that the prevalence of early childbirth in Maharashtra is higher than that in Uttar Pradesh (since the difference was calculated as Uttar Pradesh minus

Maharashtra, yielding a negative value).

The test statistic (t = -9.7053, df = 83,861, $p < 0.001$) shows that the difference in weighted prevalence between the two states is statistically significant. Therefore, we conclude that there is a significant difference in the prevalence of early childbirth between Maharashtra and Uttar Pradesh.

Logistic Regression Model Findings

Table 2: Survey-weighted logistic regression results for early childbirth

Variable	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-0.25072	0.03430	-7.310	2.69×10^{-13}
factor(V024)27	0.39745	0.02379	16.708	$< 2 \times 10^{-16}$
factor(V025)2	0.07703	0.02823	2.728	0.00637
factor(V106)1	-0.01512	0.02830	-0.534	0.59321
factor(V106)2	-0.49899	0.02312	-21.584	$< 2 \times 10^{-16}$
factor(V106)3	-2.10046	0.04952	-42.414	$< 2 \times 10^{-16}$
factor(V190)2	0.07660	0.02520	3.039	0.00237
factor(V190)3	0.14849	0.02769	5.362	8.27×10^{-8}
factor(V190)4	0.03407	0.03162	1.078	0.28119
factor(V190)5	-0.14810	0.03894	-3.804	0.00014
factor(V130)2	-0.08218	0.02920	-2.814	0.00489
factor(V130)3	-0.23091	0.25851	-0.893	0.37173
factor(V130)4	0.05772	0.28115	0.205	0.83734
factor(V130)5	-0.09887	0.07499	-1.318	0.18740
factor(V130)6	-1.00301	0.28187	-3.558	0.00037
factor(V130)8	0.49252	0.55352	0.890	0.37358
factor(V130)9	-0.35717	0.51327	-0.696	0.48651
factor(V130)96	0.69449	0.38616	1.798	0.07211
factor(V131)992	0.10239	0.04934	2.075	0.03798
factor(V131)993	0.10827	0.07606	1.423	0.15460
factor(V131)998	-0.13808	0.30999	-0.445	0.65601

The survey-weighted logistic regression model examined the association between early childbirth (dependent variable: `early_birth`, coded 1 if first birth before age 20 and 0 otherwise) and socio-demographic predictors such as state, residence, education, wealth, religion, and caste. Women from Maharashtra (V024 = 27) had significantly higher odds of early childbirth compared to those from Uttar Pradesh (reference category), as indicated by a positive and highly significant coefficient ($\beta = 0.397$, $p < 0.001$). Rural women (V025 = 2) also showed higher odds relative to urban women. Education (V106) was strongly negatively associated with early childbirth, with secondary and higher education levels showing substantially lower odds ($p < 0.001$). Women from poorer wealth quintiles (V190) were more likely to have early births than those from richer groups. Religion (V130) and caste (V131) displayed weaker and less consistent associations. Overall, the results indicate that lower education, lower wealth, rural residence, and being from Maharashtra are key risk factors for early childbirth among women in the sample.

5.5 Discussion

This study adds to the literature by (i) providing a state-level comparison of early childbirth using survey-weighted estimates, ensuring nationally representative findings, (ii) jointly modeling socio-demographic factors through a survey-adjusted logistic regression framework that accounts for complex sampling design, and (iii) identifying that Maharashtra, despite its higher development indicators, exhibits greater early childbirth prevalence than Uttar Pradesh—highlighting the need to look beyond economic development and focus on targeted social interventions.

One possible reason might be:When we define “early childbirth” as having the first birth before age 20, we’re essentially asking:“By the time a woman reaches age 20, has she already had a child?” Now, if we include girls or young women below 20 (say 15–19 years old) in the analysis,some of them might not have had a child yet,but they still could before turning 20.So, treating them as “no early childbirth” is misleading — they haven’t finished the risk period yet.

```
> sum(UP$V013 == 1, na.rm = TRUE)
[1] 19844
> sum(Maha$V013 == 1, na.rm = TRUE)
[1] 4929
```

Here, `V013 = 1` corresponds to women in the age group 15-19 years.

5.6 Conclusion

The study found that the prevalence of early childbirth is significantly higher in Maharashtra (41.9%) than in Uttar Pradesh (37.0%), indicating that economic development alone does not ensure better reproductive outcomes. Education proved to be the strongest protective factor, as women with higher education levels were far less likely to experience early childbirth. Wealth and urban residence also lowered the risk, underscoring the importance of socio-economic empowerment. The results underscore the multidimensional nature of early childbirth, where educational empowerment and socio-economic upliftment play a more decisive role than regional development alone. These insights can inform policy interventions aimed at reducing early childbirth through improvements in female education, rural health infrastructure, and community-level awareness.

6 Research Question 3:

6.1 Research objective

Infant Mortality rate is different in different states. One reason for this is obviously population, but the other thing that might be is that after birth within 3 days if food other than breast milk is given to the child like (honey,glucose water,fruit juice,JANAM GHUTTI etc. then it might increase infant mortality rate. Also this type of carelessness may occur due to lack of education of the mother. In this study our main aim is to find if there is a significant relation between these three thing - Infant mortality rate , food given other than breast milk within some days of birth and mother’s education.

6.2 Research hypothesis

Infant mortality rate , food given other than breast milk within some days of birth and mother’s education level are significantly correlated.

6.3 Methodology

The analysis uses data from the National Family Health Survey (NFHS–5), which follows a complex survey design. To ensure unbiased and representative estimates, the survey design information was incorporated using the `survey` package in R. The following elements of the design were applied:

- "B5$01","B5$02",... to "B5$20" indicates whether i th child is alive or death. Here 1 indicates alive , o indicates death , NA indictes she does not have i th child. (For i = 1,2,..,20)
- "B6$01","B6$02",... to "B6$20" indicates when her i th child died. Here 100 represents died on the day of born, 101 represents died in day 1, 102 represents on day 2,....,130 represents died on day 30. Also 201 represents died on month 1, 202 represents died on month 2,...,212 represents died on month 12. Finally 301 represents died on 1 years,3012 represents on 2 years,303 represents on 3 years and so on.. Any values other than these are considered as wrong input or not applicable values.
- "M55$1",..."M55$6" indicates whether food other than breast milk is given or not to the i th child. 1 indicates yes , 0 indicates no.
- ("M55A$1",..."M55A$6");.....("M55O$1",..."M55O$6");("M55X$1",..."M55X$6") indicates names of food item other than breast milk is given or not to the i th child. 1 indicates yes , 0 indicates no. The code name of the food items are as follows:
  - A - milk (other than breast milk)
  - B - Plain water
  - C - Sugar or glucose water.
  - D - Gripe water
  - E - Sugar or salt solution
  - F - Fruit Juice
  - G - Infant Formula
  - H - tea or infusions
  - I - Honey
  - J - Coffee
  - K - JANAM GHUTTI
  - L,M,M,O - country specific/traditional food
  - X - Anything other food item.
- "V107" indicates highest year of education.

Calculation of NMR,PNMR,IMR

- For 1st child we calculate the following:
  - we extract the "B5$01" and then after removing the "NA"s we sum the total colunm which gives us the total number of living child (l1)
  - Then extract the "B6$01". Similarly omitting all the "NA"s.

  - For calculating neonatal deaths of 1st child, we calculate number of children died within 28 days using
    NM1 = length(d1_clean[d1_clean <= 128])
  - For calculating post neonatal deaths of 1st child, we calculate number of children died between 28 days to 1 year using
    PNM1 = length(d1_clean[(d1_clean <= 212 & d1_clean >=200) | d1_clean ==129 | d1_clean == 130])
  - For calculating infant deaths of 1st child, we calculate number of children died within 1 year using
    IM1 = length(d1_clean[d1_clean <= 212])

- Similarly we calculate (NM2,...NM20);(PNM2,...,PNM20);(IM2,...,IM20) for 2nd,..,20 th child respectively.

- So, total Neo mortal deaths, NM = NM1+NM2+..+NM20
  total Post Neo mortal deaths, PNM = PNM1+PNM2+..+PNM20
  total Infant deaths, IM = IM1+IM2+..+IM20
  total living children, l = l1+l2+..+l20

- So, the NMR = NM/l*1000
  the PNMR = PNM/l*1000
  the IMR = IM/l*1000

  After computing NMR,PNMR,IMR for both the states UP and Maharastra we plot the values in a graph and compare those.

Relation between Infant mortality and foods other than breast milk given within few days of birth

Here our objective is to find if there is any significant relation between Infant mortality and foods other than breast milk given within few days of birth. To do that we proceed as following :-

- Find the numbers of food item given to child within 3 days of birth for each individual food item.

- Among them we pick the most 5 consumed food item and then find we find correlation with these with the child whether alive or dead.

- We done this for the 1st child only as number of other child and relevant food consumed by them - the number is relevantly very low.

Relation between infant mortality rate , food given other than breast milk within some days of birth and mother's education level

Here our objective is to find if there is any significant relation between infant mortality rate , food given other than breast milk within some days of birth and mother's education level. To do that we proceed as following :-

- We take the column "B$601", "M55$1" , "V107" and after removing the "NA"s we compute partial correlation between them.

Logistic regression fitting

Finally we examine the effect of feeding practices and mother's education on child survival by fitting a logistic regreesion.

6.4 Salient Findings

First let us see the value of NMR,PNMR,IMR for UP and Maharashtra separately:-

```
> NMR_UP
[1] 43.74661
> PNMR_UP
[1] 18.32515
> IMR_UP
[1] 62.09562
```

Figure 4: UP_NMR__PNMR__IMR

```
> NMR_Maharastra
[1] 23.32889
> PNMR_Maharastra
[1] 8.45832
> IMR_Maharastra
[1] 31.78721
```

Figure 5: Maharashtra_NMR__PNMR__IMR

Now compare these results by plotting in a bar plot : -

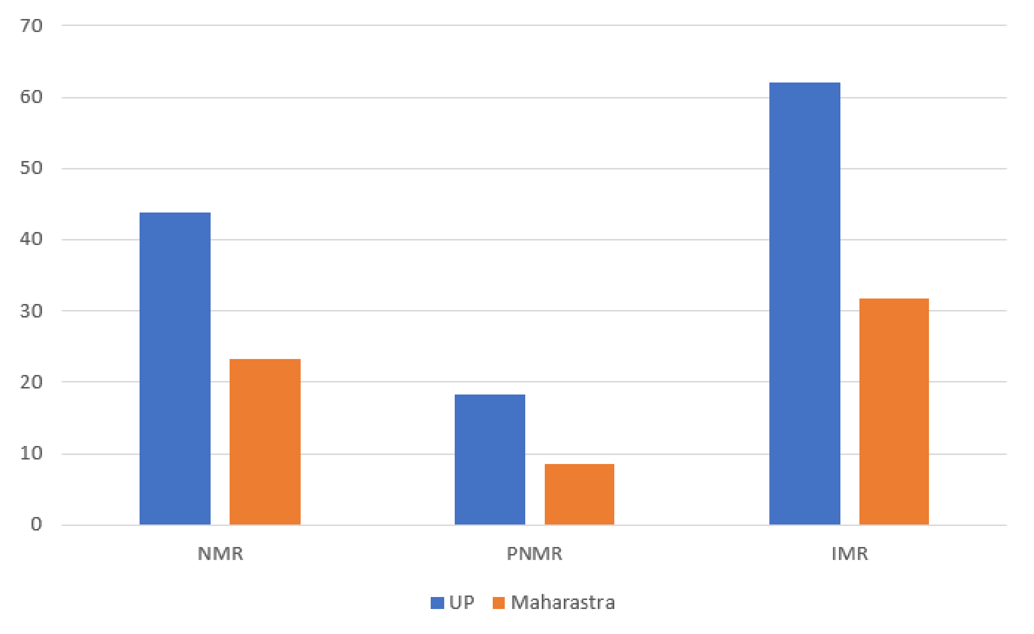


Figure 6: Comparing _NMR__PNMR__IMR

So in all of the infant death rates UP has higher rates than Maharashtra. This may have many reasons but from the data what suggests is that proper caring of the child after birth may one reason.A newborn's stomach and intestines are still developing. They cannot digest or absorb nutrients from solid or even semisolid foods efficiently.Introducing other foods too early causes indigestion, diarrhea, and nutrient loss. So ,giving food other than breast milk may be one cause of this infant mortality rate. So, we check correlation between death and food consumed (top 5 food consumed for each state) and got the following results :-

```
> corA_UP
          M55A$1
B5$01 0.01955322
> corB_UP
            M55B$1
B5$01 -0.004541881
> corH_UP
          M55H$1
B5$01 0.01581719
> corI_UP
          M55I$1
B5$01 0.01541789
> corK_UP
         M55K$1
B5$01 0.0150255
```

Figure 7: Correlation of Child alive or death and food consumed after birth for UP

```
> corA_Maharashtra
         M55A$1
B5$01 0.0212993
> corB_Maharashtra
           M55B$1
B5$01 -0.01288785
> corC_Maharashtra
            M55C$1
B5$01 -0.001001095
> corG_Maharashtra
            M55G$1
B5$01 -0.001001095
> corI_Maharashtra
           M55I$1
B5$01 0.009430586
```

Figure 8: Correlation of Child alive or death and food consumed after birth for Maharashtra

To find the above correlation we find our necessary variables by counting which food items are mostly consumed in the two states. We remove the other variables because the number of other variable is too low that do not have significant effect in determining the relation with death. Also we do this only for the first child as number of other children are too low or negligible.

We find the most consumed food by children in UP are -

- A - milk (other than breast milk)
- B - Plain water
- H - tea or infusions

- I - Honey
- K - JANAM GHUTTI

We find the most consumed food by children in Maharashtra are -

- A - milk (other than breast milk)
- B - Plain water
- C - Sugar or glucose water.
- G - Infant Formula
- I - Honey

Now after calculating correlation we observe that in all the cases the correlation is too low or negative in also some cases. So this implies that relation between the child is alive or not and food other than breast milk given within 3 days is negatively correlated. This means food consumed other than breast milk within 3 days of birth may cause infant mortality and hence can increase infant mortality rate.

After this the natural question which arise is that why the parents give these type of food to their new born childrens ? This occurs may be due to mothers with low or no formal education often don't know that exclusive breastfeeding (only breast milk for 6 months) is scientifically recommended by WHO and UNICEF.They may believe that babies need water, cow's milk, or semi-solid food early to grow properly.This misunderstanding directly leads to unsafe feeding and higher infant illness. So this encourages us to test if there is a significant relation between Infant mortality rate , food given other than breast milk within some days of birth and mother's education level. We obtain the following results : -

```
> cat("Partial correlation between B6_01 and V107 controlling for M55_1:\n")
Partial correlation between B6_01 and V107 controlling for M55_1:
> print(pcor1)
     estimate   p.value statistic   n gp  Method
1 -0.05944446 0.2898364 -1.060254 320  1 pearson
>
> cat("\nPartial correlation between B6_01 and M55_1 controlling for V107:\n")

Partial correlation between B6_01 and M55_1 controlling for V107:
> print(pcor2)
    estimate   p.value statistic   n gp  Method
1 0.07211217 0.1989385  1.287272 320  1 pearson
```

Figure 9: Partial calculation_UP

```
Partial correlation between B6_01 and V107 controlling for M55_1:
> print(pcor1)
    estimate   p.value  statistic  n gp  Method
1 -0.1064682 0.3911573 -0.8632809 68  1 pearson
>
> cat("\nPartial correlation between B6_01 and M55_1 controlling for V107:\n")

Partial correlation between B6_01 and M55_1 controlling for V107:
> print(pcor2)
    estimate   p.value statistic  n gp  Method
1 0.03117256 0.8022646 0.2514434 68  1 pearson
```

Figure 10: Partial calculation_Maharashtra

So here also we find that partial correlation between deaths of infants and education level of mother given food other than breast milk given within 3 days is negatively correlated which means mother's education level has significant impact on reducing infant mortality. If a mother has good education then she must know about her child's proper diet. Also good health means reduction in infant mortality. Same kind of results holds for partial correlation between deaths of infants and given food other than breast milk given within 3 days given education level of mother.

Fitting logistic regression

Below is the model summary and interpretation after fitting the model:-

Model Summary

- Number of observations: 17,134
- Null deviance: 3181.5 Residual deviance: 3168.8
- AIC: 3174.8
- Model accuracy: 98.1%

Regression Coefficients

Variable	Estimate	Std. Error	z value	p-value	Odds Ratio	95% CI (OR)
Intercept	3.9196	0.1657	23.65	<0.001	50.38	36.56–70.03
M55$1	0.5200	0.1550	3.36	0.0008	1.68	1.25–2.30
V107	-0.0136	0.0350	-0.39	0.699	0.99	0.92–1.06

Interpretation

Model Fit and Accuracy

The model explains child survival fairly well, with a high accuracy of 98.1% in classifying whether a child is alive or dead. The small difference between the null deviance (3181.5) and residual deviance (3168.8) suggests that the predictors slightly improve the model fit compared to a model with no predictors.

Effect of Feeding Practices (M55$1)

The coefficient for `M55$1` is positive (β = 0.52) and statistically significant (p = 0.0008). The odds ratio of 1.68 indicates that children who were given food or liquids other than breast milk are 1.68 times more likely to be alive compared to those who were exclusively breastfed, holding education constant. This suggests that appropriate complementary feeding contributes to better child survival, though causality should be interpreted carefully.

Effect of Mother's Education (V107)

The coefficient for `V107` is negative (β = -0.0136) but not statistically significant (p = 0.699). The odds ratio (0.99) shows almost no change in the odds of survival with one additional level of education, after controlling for feeding practices. Hence, in this dataset, mother's education level does not have a significant independent effect on child survival.

Overall summary

Feeding practices are significantly associated with higher child survival, whereas mother's education level does not show a significant effect. The model performs well overall but may be influenced by the imbalance in alive versus dead cases. Including additional demographic and socio-economic variables in future models may improve predictive performance.

6.5 Discusssion

Reasons for Higher Infant Mortality in Uttar Pradesh than in Maharashtra

1. Healthcare Infrastructure and Access

Uttar Pradesh has comparatively weaker healthcare infrastructure, especially in rural areas, with fewer hospitals, health sub-centers, and trained medical personnel. Maharashtra, being more urbanized and economically advanced, ensures better access to institutional deliveries, neonatal care units, and emergency obstetric services.

2. Maternal Health and Nutrition

Maternal malnutrition, anemia, and inadequate antenatal care are more prevalent in Uttar Pradesh. Poor maternal health directly affects the infant's birth weight and survival chances. Maharashtra performs better in implementing maternal and child health programs, improving overall outcomes.

3. Socioeconomic Factors

Higher poverty, lower literacy, and weaker female empowerment in Uttar Pradesh contribute to poor child health outcomes. Educated mothers are more likely to seek medical assistance, maintain hygiene, and adopt family planning. Maharashtra benefits from better living standards and awareness levels.

4. Sanitation and Safe Water

Unsafe drinking water, open defecation, and poor sanitation in many parts of Uttar Pradesh lead to diarrheal and infectious diseases, which are major causes of infant deaths. Maharashtra has improved sanitation and water supply facilities, reducing disease incidence.

5. Cultural and Behavioral Factors

Traditional beliefs, early marriage, and closely spaced pregnancies are more common in Uttar Pradesh, increasing risks to both mother and child. Maharashtra has better health-seeking behavior and wider acceptance of modern healthcare practices.

6. Implementation of Government Programs

The implementation of national health programs such as the National Rural Health Mission (NRHM) and the Reproductive, Maternal, Newborn, Child and Adolescent Health (RMNCH+A) framework faces challenges in Uttar Pradesh due to resource and management gaps. Maharashtra demonstrates stronger governance, better monitoring, and more effective service delivery.

7. Urbanization and Infrastructure

Maharashtra's higher degree of urbanization allows easier access to healthcare and education, while Uttar Pradesh's predominantly rural population faces difficulties in receiving timely and quality healthcare services.

6.6 Conclusion

In summary, the higher infant mortality rate in Uttar Pradesh results from poor healthcare access, lower maternal education and nutrition, inadequate sanitation, and weaker implementation of health programs.

To overcome this situation, it is essential to strengthen primary healthcare infrastructure, ensure timely maternal and child health services, and improve access to institutional deliveries and neonatal care. Increasing female education and awareness about hygiene, nutrition, and immunization can significantly reduce infant deaths. Effective implementation and monitoring of government schemes such as the National Health Mission and POSHAN Abhiyaan are also crucial.

Improving sanitation, safe drinking water, and reducing poverty through community-based interventions can further enhance child survival. A combined effort of government, community participation, and awareness campaigns can help Uttar Pradesh achieve lower infant mortality and better overall health outcomes, similar to Maharashtra.

7 Research Question 4:

7.1 Research Objective

The study aims to examine whether residence factors explain the variation in fertility levels. Specifically, we test whether a woman's residence has a statistically significant effect on fertility.

7.2 Research Hypothesis

Hypothesis 1

We are going to fit a regression model:

$$Y = \beta_0 + \beta_1 Rural + \epsilon \tag{1}$$

Where

- Y=Total no of children born.
- Rural=2 if not urban else Rural=1.

- Null hypothesis (H_0):

$$\beta_1 = 0$$

- Alternative hypothesis (H_1):

$$\beta_1 \neq 0$$

Comparison of Mean Fertility by Residence

To examine whether the mean number of children ever born (`V201`) differs between urban and rural women, a two-sample t-test was conducted. The analysis was performed separately for Maharashtra and Uttar Pradesh using the complex survey design of NFHS–25.

Let μ_U denote the mean number of children ever born among urban women and μ_R denote the corresponding mean among rural women.

Hypothesis 2

- Null hypothesis (H_0):

$$\mu_U = \mu_R$$

  or equivalently,

$$\mu_U - \mu_R = 0$$

- Alternative hypothesis (H_1):

$$\mu_U \neq \mu_R$$

7.3 Methodology

The analysis uses data from the National Family Health Survey (NFHS–5), which follows a complex survey design. To ensure unbiased and representative estimates, the survey design information was incorporated using the `survey` package in R. Three elements of the design were applied:

- **Sampling weights** ($V005$) were used to adjust for unequal probability of selection.
- ($v025$) Type of place of residence.
- ($v201$) Total no of child birth.

The survey design was specified in R as:

```
nfhsdesign <- svydesign( id = 021, strata = 022, weights =  weight, data = dataset, nest = TRUE)
```

Survey Regression Model

Fertility (measured as number of children ever born, variable V 201) is a count variable. So a regression model is used. It is the same model described in the hypothesis part of Research Question 4. This model is fitted for both states separately.

Variable Description

Dependent Variables

- Number of children ever born (`v201`)

Independent Variables

1. **Type of place of Residence of Women** $(V025)$

Value	Label
1	Urban
2	Rural

7.4 Salient Findings

Desciptive Statistics

We have calculated the mean no of children born in urban and rural areas for both Maharashtra and UP.By observing the charts below, we can see that the fertility is higher in rural than in urban areas for Maharashtra. Also, the distribution of fertility is lower in urban than rural areas in Maharashtra.
By observing the charts below, we can see even though the fertility is higher in rural than in urban areas for Maharashtra. But,the distribution of fertility is almost same in urban and rural areas in UP.

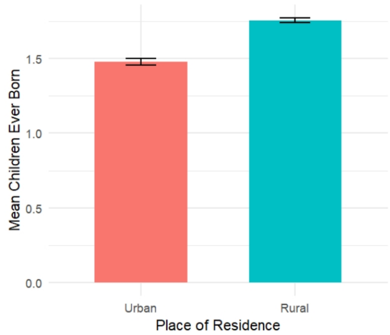


Figure 11: Maharasthra(Avg Children born by Place of Residence)

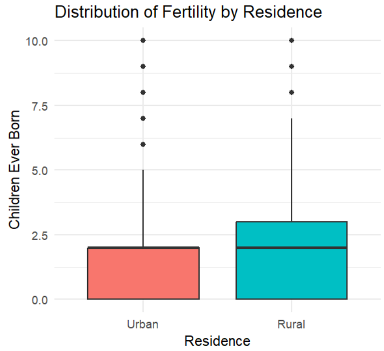


Figure 12: Maharasthra

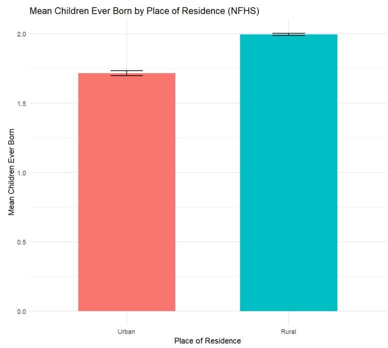


Figure 13: UP

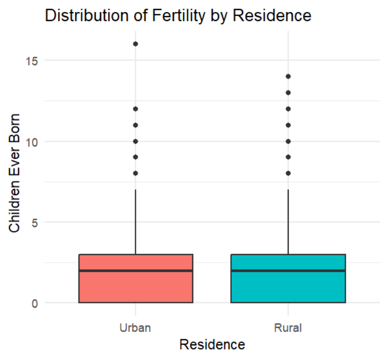


Figure 14: UP

Regression findings

From the values below, we can conclude that the coefficient of rural/urban is almost the same for both Maharashtra and UP. Since the p-value for both states is less than 0.001. So, we have enough evidence to reject $\beta_1 = 0$.

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.47808    0.02311   63.95   <2e-16 ***
residenceRural 0.27643    0.02732   10.12   <2e-16 ***
```

Figure 15: Survey Regression Output For Maharashtra

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.71621    0.01774   96.75   <2e-16
residenceRural 0.27775    0.01933   14.37   <2e-16
```

Figure 16: Survey Regression Output For UP

t-test findings

- **Maharashtra**: The two-sided t-test results are as follows:
  t = 10.117, df = 1371, p-value <0.05.
  Alternative hypothesis: true difference in mean is not equal to 0.
  95 percent confidence interval: (0.2228298 0.3300258).
  sample estimates: difference in mean 0.2764278.

- **UP**: The two-sided t-test results are as follows:
  t = 14.372, df = 3057, p-value < 2.2e-16

Alternative hypothesis: true difference in mean is not equal to 0.
95 percent confidence interval: (0.2398581 0.3156419)
sample estimates: difference in mean 0.27775

7.5 Discussion

Surprisingly, according to the above findings, if we move the UP rural women to the urban areas, it will not affect the fertility by figure 12. But the above method does affect the fertility in the case of the Maharashtra women by figure 11.

```
      residence     V201          se
Urban     Urban 1.478078 0.02311340
Rural     Rural 1.754505 0.01456976
```

Figure 17: Weighted Mean for Maharashtra

```
      residence     V201           se
Urban     Urban 1.716207 0.017737880
Rural     Rural 1.993957 0.007670438
```

Figure 18: Weighted Mean for UP

7.6 Conclusion

So the place of residence matters more for Maharashtra women than UP women. Theoretically, we can keep the fertility rate in Maharashtra under control by changing the place of residence. But it will not affect the UP women much to Figure 7. The t-test gives us enough evidence to reject that the mean fertility for urban and rural areas is not the same for both states.The conclusion from the test are as follows:

- By the regression model, we get that there is a linear relationship between V201 and the Residence Place variable.
- The distribution of fertility in UP is the same in rural and urban areas by figure 12.
- The distribution of fertility in Maharashtra is larger in rural than urban areas by figure 11.
- The two-sided t-test concluded both states do not have same fertility rate in rural and urban areas.

8 Research Question 5:

8.1 Introduction

The western region of Uttar Pradesh benefits greatly from its proximity to the National Capital Region (NCR), which has driven the development of better infrastructure such as roads, expressways, and industrial clusters. This strategic advantage has attracted investment, encouraged industrialization, and created employment opportunities, leading to higher income levels and faster economic growth.

In contrast, the eastern and more backward zones of the state have suffered from historical neglect, inadequate infrastructure, and limited economic diversification. These areas remain heavily dependent on primary agriculture, with smaller and more vulnerable landholdings that constrain productivity and income generation. As a result, Uttar Pradesh exhibits a dualistic economic geography—where the western part has evolved into a richer, more urbanized, and cosmopolitan belt, while the eastern region continues to lag behind as a predominantly agrarian and less developed area.

Similarly, Western Maharashtra, particularly the coastal belt encompassing cities like Mumbai and Pune, is highly industrialized and hosts a large share of the state's services sector jobs, supported by superior infrastructure and strong economic linkages. In contrast, the interior regions such as Marathwada, Vidarbha, and North Maharashtra remain largely agrarian, face frequent droughts, and have limited presence of high-value industries, leading to slower consumption and income growth.

The hinterland or less-accessible interior districts have consequently lagged behind the prosperous coastal and metropolitan areas. This disparity is driven by the concentration of capital, infrastructure, and services in a few western districts, while the eastern and interior zones suffer from less favorable climatic conditions, inadequate irrigation, and lower industrial investment. Additionally, a pronounced rural–urban divide exacerbates the imbalance, as rural districts fail to keep pace with the rapid economic expansion of urban centers. Together, these regional and structural inequalities have amplified Maharashtra's internal economic divide, creating a stark contrast between its developed western core and its lagging interior regions.

8.2 Research Hypothesis

Although Uttar Pradesh and Maharashtra both exhibit regional income inequality, the **spatial variance of income distribution among districts is greater in Uttar Pradesh** than in Maharashtra.

The following maps illustrate income distributions as per 2019–20 data.

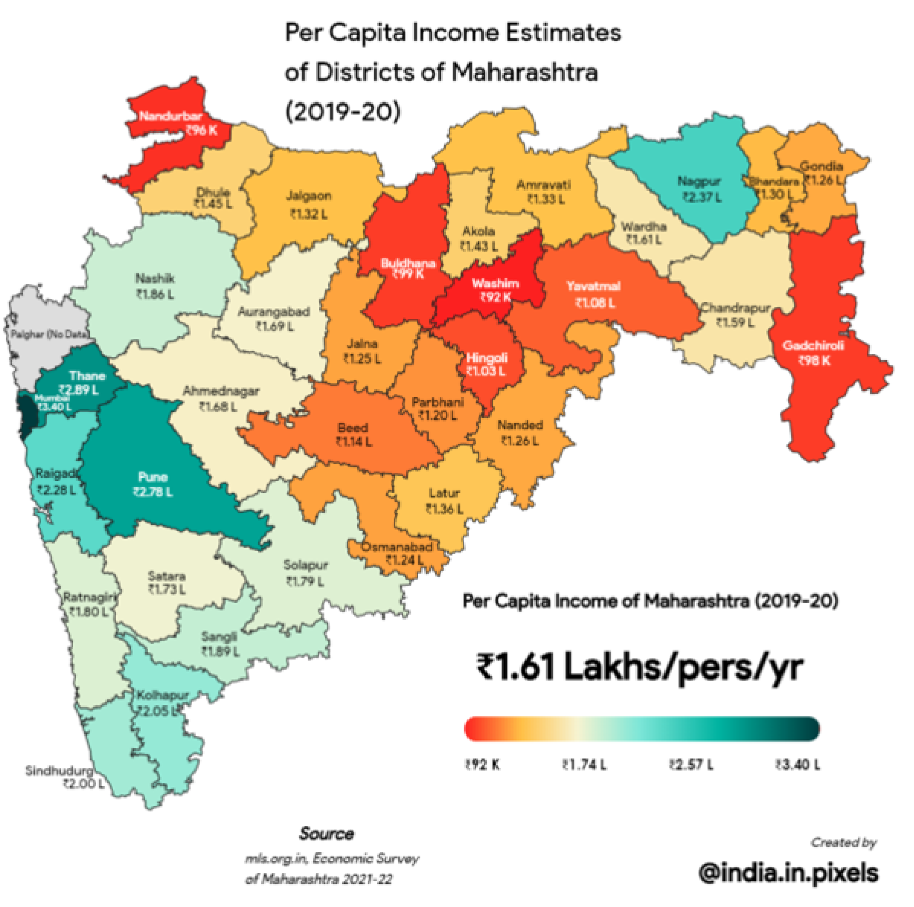


Figure 19

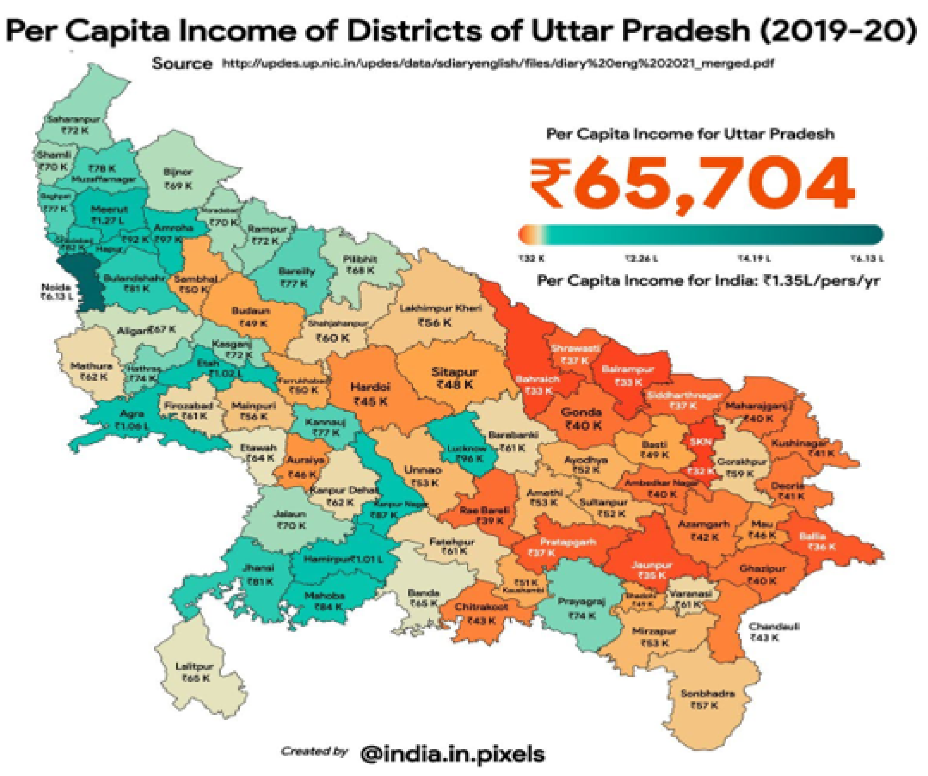


Figure 20

8.2.1 STRUCTURE OF THE DATA

For both the states, we enlist the districts in alphabetic order alongside their corresponding latitude and longitude which captures the spatial information and also the average income of the districts.

8.3 Methodology

In order to calculate the spatial variance, we make use of the **variogram**. A variogram is a fundamental tool in **geostatistics** used to describe the degree of **spatial dependence** between sample points in a spatial dataset. It quantifies how similar (or dissimilar) values are as a function of the distance between them.

Experimental Variogram

The experimental variogram is defined as:

$$\gamma(h) = \frac{1}{2N(h)} \sum_{i=1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2$$

- $\gamma(h)$: semivariance for lag distance h
- $N(h)$: number of point pairs separated by distance h
- $Z(x_i)$: observed value at location x_i
- $Z(x_i + h)$: observed value at distance h from x_i

Interpretation

- When $h = 0$: $\gamma(0) = 0$ (no difference at the same location)

- As h increases, $\gamma(h)$ typically increases — values farther apart are less correlated.
- Eventually, $\gamma(h)$ reaches a plateau called the **sill**, beyond which spatial dependence disappears.
- The distance at which this happens is the **range**.

Components of a Variogram

- **Nugget:** Variance at zero distance (measurement error or microscale variability).
- **Sill:** The total variance when points are no longer spatially correlated.
- **Range:** The distance beyond which values become spatially independent.

Calculating Spatial Variance

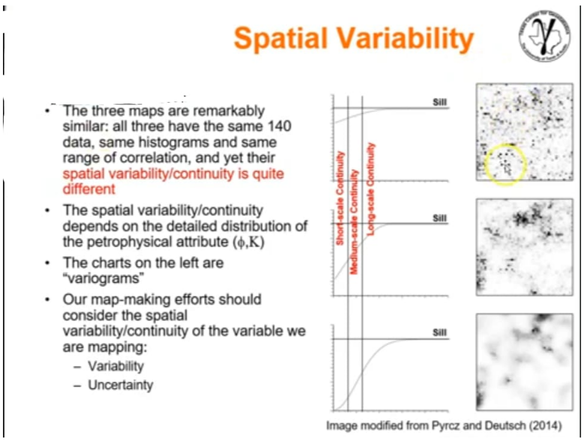
The spatial variance (total variability) of the dataset can be inferred from the sill of the variogram.

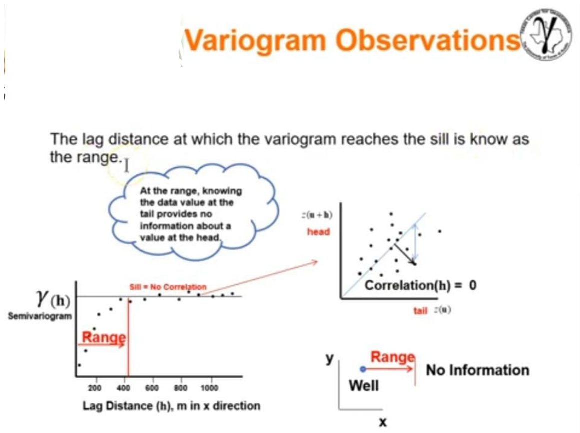
•The sill approximates the overall variance of the spatial process, i.e.,

$$\text{Spatial Variance} \approx \text{Sill of the Variogram}$$

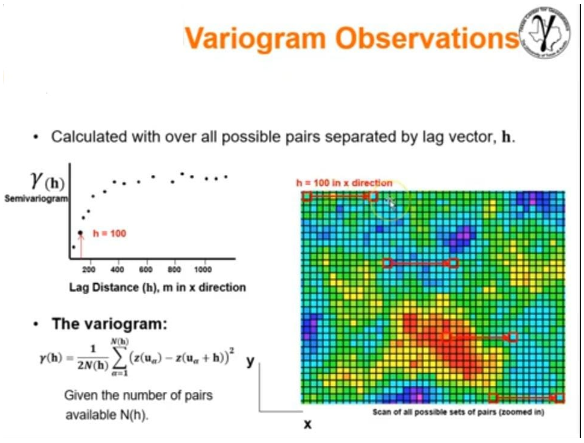
One important caveat that may even put the interpretation of our results in doubt is that the method of variogram works only with stationary spatial data : having constant mean and variance which is clearly untrue for both our calculations as we have a gradient from left to right. Therefore, we take the square root of the data to make it stationary.

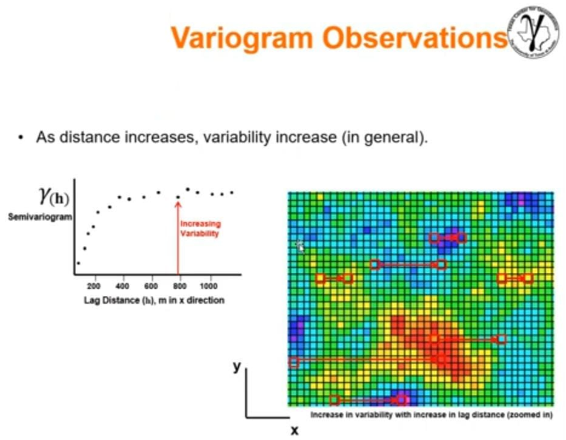
8.4 Interpretation of a Variogram





Based on the skgstat library, the Variogram function parameters configure the spatial analysis. coordinates are the (x, y) or (lat, lon) locations of your data points, while values are the actual measurements at those locations. maxlag defines the maximum distance to consider for the analysis which is 1000 kilometres in our case, and n_lags specifies how many distance intervals (bins) to divide this range into. bin_func='uniform' ensures these bins have equal width which is reasonable since the locations of our districts is uniform over the state. The estimator='matheron' selects the classic Matheron formula (mentioned earlier) to calculate the experimental semivariance. metric='haversine' is crucial for geographical data, as it correctly calculates the great-circle distance between latitude/longitude pairs on a sphere. Finally, model='spherical' specifies the theoretical model (describing the sill, range, and nugget) that will be fitted and it is used as it is the most





standard and widely used model for a variety of situations. An example calculation of a variogram with simulated data is given below.

8.5 Salient Findings

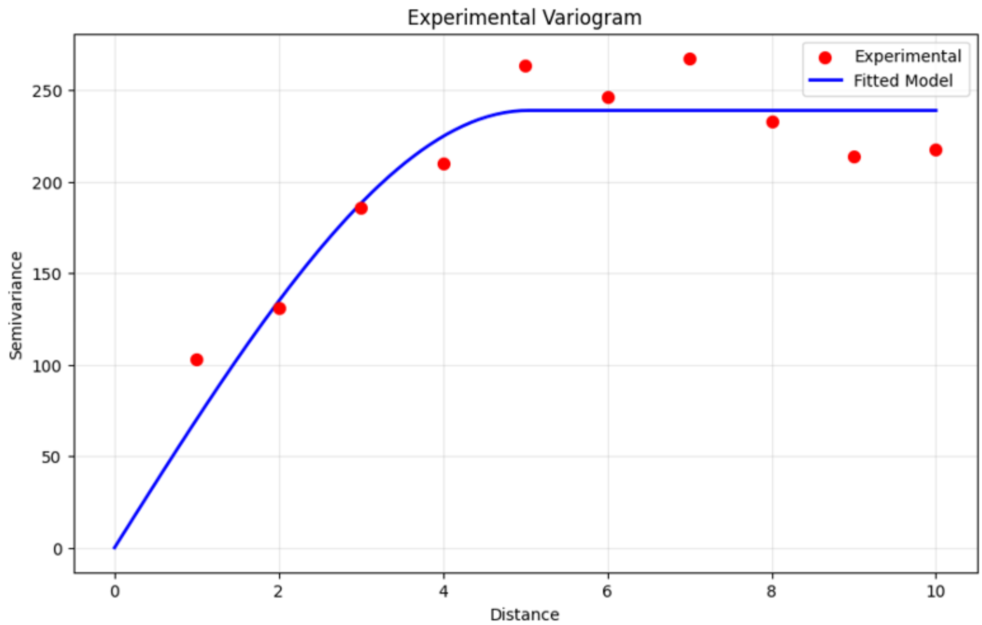
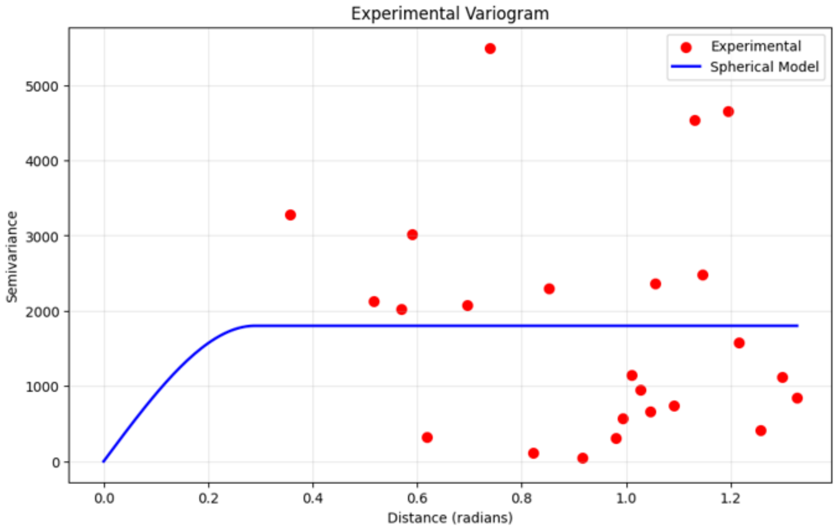


Figure 21

The initial part of the variogram tells about short range variability, the next intermediate part about medium range variability and the value attained at sill after the range is indicative of the long range variability.

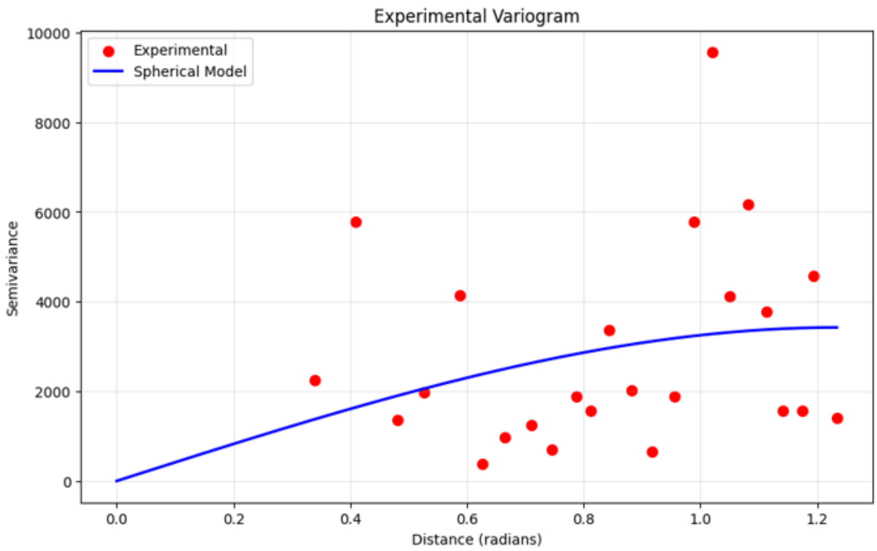
For Maharashtra square root of income against the coordinate of each of the 36 districts, we obtain the following Variogram.




```
Nugget: 0.289
Sill: 1802.005
Range (radians): 0.000000
Range ≈ 0.00 km
```

Figure 22

For UP,our Variogram is :




```
Nugget: 1.234
Sill: 3423.089
Range (radians): 0.000000
Range ≈ 0.00 km
```

Figure 23

8.6 Discussion

Reasons for Higher Spatial Variance of Income Distribution in Uttar Pradesh than in Maharashtra is :

Although Uttar Pradesh and Maharashtra both exhibit regional income inequality, the spatial variance of income distribution among districts is greater in Uttar Pradesh. This is due to the following factors:

1. Uneven Economic Development

Maharashtra has multiple growth centers such as Mumbai, Pune, Nashik, and Nagpur, which promote balanced industrial and service sector expansion. In contrast, economic development in Uttar Pradesh is concentrated in a few urban centers like Noida, Lucknow, and Kanpur, while large parts of eastern and central regions remain predominantly agricultural and underdeveloped.

2. Industrial and Urban Concentration

The industrial corridor in Maharashtra, particularly the Mumbai–Pune–Nashik belt, supports the diffusion of income-generating activities. Uttar Pradesh lacks similar industrial linkages, resulting in localized growth and higher inter-district disparities.

3. Agriculture versus Diversified Economy

Uttar Pradesh's economy is heavily dependent on low-productivity agriculture, whereas Maharashtra's diversified base in manufacturing, services, and agro-industry reduces income differences between districts.

4. Human Capital and Education

There are significant differences in literacy and skill levels across Uttar Pradesh, with the western region being more developed than the eastern part. Maharashtra, on the other hand, exhibits more uniform educational development, contributing to relatively balanced income levels.

5. Infrastructure and Investment Disparities

Infrastructure facilities such as roads, electricity, and communication networks are unevenly distributed in Uttar Pradesh. Maharashtra's consistent investment in connectivity and industrial infrastructure promotes more balanced regional growth.

6. Governance and Policy Effectiveness

Maharashtra benefits from more stable governance and effective local institutions, leading to better implementation of regional development policies. Uttar Pradesh faces administrative challenges that contribute to uneven development outcomes.

8.7 Conclusion

Since the nugget values are negligible compared to the sill values, the sill approximates the overall **variance of the spatial process**,i.e.,

$$\text{Spatial Variance} \approx \text{Sill of the Variogram}.$$

Thus the spatial variance of income is much higher for UP than it is for Maharashtra.

9 Work allocation

Roll.No	Name of member	Contribution
BS2304	ADITY BANERJEE	QUESTION 2
BS2310	ANJAN SADHUKHAN	QUESTION 1
BS2322	AYUSH BARAN SEN	QUESTION 4
BS2337	RAHUL GANGULY	QUESTION 3
BS2356	VINAYAK TRIPATHI	QUESTION 5

Table 3: Work allocation